Gesture Recognition

10.1、Introduction

MediaPipe is a data stream processing machine learning application development framework developed and open-sourced by Google. It is a graph-based data processing pipeline for building with various forms of data sources such as video, audio, sensor data, and any time series data.

MediaPipe is cross-platform and can run on embedded platforms (Raspberry Pi, etc.), mobile devices (iOS and Android), workstations and servers, and supports mobile GPU acceleration. MediaPipe provides cross-platform, customizable ML solutions for real-time and streaming media.

MediaPipe's core framework is implemented in C++ and provides support for Java and Objective C. The main concepts of MediaPipe include Packet, Stream, Calculator, Graph, and Subgraph.

MediaPipe features:

- End-to-end acceleration: Built-in fast ML inference and processing accelerates even on ordinary hardware.
- Build once, deploy anywhere: Unified solutions work on Android, iOS, desktop/cloud, web, and IoT.
- Ready-to-use solutions: Cutting-edge ML solutions that showcase all framework features.
- Free and open source: Framework and solutions under Apache 2.0, fully extensible and customizable.

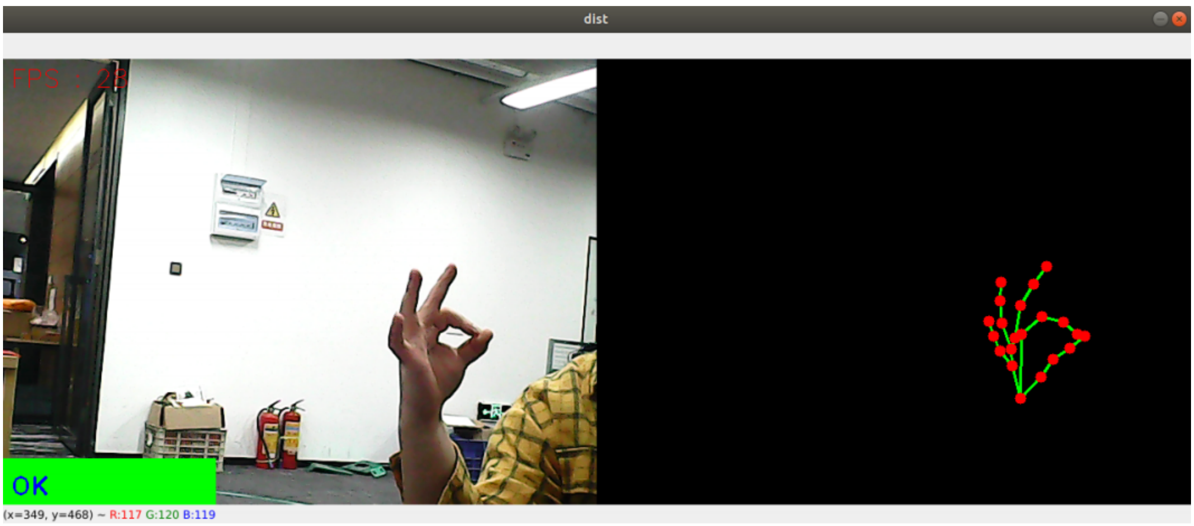
10.2、Gesture Recognition

Hand gesture recognition designed for the right hand, which can accurately recognize when specific conditions are met. Recognizable gestures include: [Zero, One, Two, Three, Four, Five, Six, Seven, Eight, Ok, Rock, Thumb_up (like), Thumb_down (thumb down), Heart_single (single hand heart)], a total of 14 types.

10.2.1、Startup

Enter in terminal,

```
ros2 run dofbot_mediapipe 10_GestureRecognition
```



10.2.、Source Code

Source location: /home/yahboom/dofbot_ws/src/dofbot_mediapipe/dofbot_mediapipe/10_GestureRecognition.py

```
#!/usr/bin/env python3
# encoding: utf-8

import math
import time
import cv2 as cv
import numpy as np
import mediapipe as mp
import rclpy
from rclpy.node import Node
from sensor_msgs.msg import Image
from cv_bridge import CvBridge

class HandDetector:
    def __init__(self, mode=False, maxHands=2, detectorCon=0.5, trackCon=0.5):
        self.tipIds = [4, 8, 12, 16, 20]
        self.mpHand = mp.solutions.hands
        self.mpDraw = mp.solutions.drawing_utils
        self.hands = self.mpHand.Hands(
            static_image_mode=mode,
            max_num_hands=maxHands,
            min_detection_confidence=detectorCon,
            min_tracking_confidence=trackCon
        )
        self.lmList = []
        self.lmDrawSpec = mp.solutions.drawing_utils.DrawingSpec(color=(0, 0,
255), thickness=-1, circle_radius=6)
        self.drawSpec = mp.solutions.drawing_utils.DrawingSpec(color=(0, 255,
0), thickness=2, circle_radius=2)

    def get_dist(self, point1, point2):
        x1, y1 = point1
        x2, y2 = point2
        return abs(math.sqrt(math.pow(abs(y1 - y2), 2) + math.pow(abs(x1 - x2),
2)))

    def calc_angle(self, pt1, pt2, pt3):
        point1 = self.lmList[pt1][1], self.lmList[pt1][2]
        point2 = self.lmList[pt2][1], self.lmList[pt2][2]
        point3 = self.lmList[pt3][1], self.lmList[pt3][2]
        a = self.get_dist(point1, point2)
        b = self.get_dist(point2, point3)
        c = self.get_dist(point1, point3)
        try:
            radian = math.acos((math.pow(a, 2) + math.pow(b, 2) - math.pow(c,
2)) / (2 * a * b))
            angle = radian / math.pi * 180
        except:
            angle = 0
        return abs(angle)

    def findHands(self, frame, draw=True):
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        self.lmList = []
        img = np.zeros(frame.shape, np.uint8)
        img_RGB = cv.cvtColor(frame, cv.COLOR_BGR2RGB)
        self.results = self.hands.process(img_RGB)
        if self.results.multi_hand_landmarks:
            for i in range(len(self.results.multi_hand_landmarks)):
                if draw: self.mpDraw.draw_landmarks(frame,
self.results.multi_hand_landmarks[i], self.mpHand.HAND_CONNECTIONS,
self.lmDrawSpec, self.drawSpec)
                self.mpDraw.draw_landmarks(img,
self.results.multi_hand_landmarks[i], self.mpHand.HAND_CONNECTIONS,
self.lmDrawSpec, self.drawSpec)
                for id, lm in
enumerate(self.results.multi_hand_landmarks[i].landmark):
                    h, w, c = frame.shape
                    cx, cy = int(lm.x * w), int(lm.y * h)
                    self.lmList.append([id, cx, cy])
        return frame, img

    def frame_combine(self, frame, src):
        if len(frame.shape) == 3:
            frameH, frameW = frame.shape[:2]
            srcH, srcW = src.shape[:2]
            dst = np.zeros((max(frameH, srcH), frameW + srcW, 3), np.uint8)
            dst[:, :frameW] = frame[:, :]
            dst[:, frameW:] = src[:, :]
        else:
            src = cv.cvtColor(src, cv.COLOR_BGR2GRAY)
            frameH, frameW = frame.shape[:2]
            imgH, imgW = src.shape[:2]
            dst = np.zeros((frameH, frameW + imgW), np.uint8)
            dst[:, :frameW] = frame[:, :]
            dst[:, frameW:] = src[:, :]
        return dst

    def fingersUp(self):
        fingers = []
        # Thumb
        if (self.calc_angle(self.tipIds[0], self.tipIds[0] - 1, self.tipIds[0] -
2) > 150.0) and (
                self.calc_angle(self.tipIds[0] - 1, self.tipIds[0] - 2,
self.tipIds[0] - 3) > 150.0):
            fingers.append(1)
        else:
            fingers.append(0)
        # 4 fingers
        for id in range(1, 5):
            if self.lmList[self.tipIds[id]][2] < self.lmList[self.tipIds[id] -
2][2]:
                fingers.append(1)
            else:
                fingers.append(0)
        return fingers

    def get_gesture(self):
        gesture = ""
        fingers = self.fingersUp()
        if self.lmList[self.tipIds[0]][2] > self.lmList[self.tipIds[1]][2] and \
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                self.lmList[self.tipIds[0]][2] > self.lmList[self.tipIds[2]][2]
and \
                self.lmList[self.tipIds[0]][2] > self.lmList[self.tipIds[3]][2]
and \
                self.lmList[self.tipIds[0]][2] > self.lmList[self.tipIds[4]][2]:
            gesture = "Thumb_down"

        elif self.lmList[self.tipIds[0]][2] < self.lmList[self.tipIds[1]][2] and
\
                self.lmList[self.tipIds[0]][2] < self.lmList[self.tipIds[2]][2]
and \
                self.lmList[self.tipIds[0]][2] < self.lmList[self.tipIds[3]][2]
and \
                self.lmList[self.tipIds[0]][2] < self.lmList[self.tipIds[4]][2]
and \
                self.calc_angle(self.tipIds[1] - 1, self.tipIds[1] - 2,
self.tipIds[1] - 3) < 150.0:
            gesture = "Thumb_up"
        if fingers.count(1) == 3 or fingers.count(1) == 4:
            if fingers[0] == 1 and (
                    self.get_dist(self.lmList[4][1:], self.lmList[8][1:]) <
self.get_dist(self.lmList[4][1:], self.lmList[5][1:])
            ):
                gesture = "OK"
            elif fingers[2] == fingers[3] == 0:
                gesture = "Rock"
            elif fingers.count(1) == 3:
                gesture = "Three"
            else:
                gesture = "Four"
        elif fingers.count(1) == 0:
            gesture = "Zero"
        elif fingers.count(1) == 1:
            gesture = "One"
        elif fingers.count(1) == 2:
            if fingers[0] == 1 and fingers[4] == 1:
                gesture = "Six"
            elif fingers[0] == 1 and self.calc_angle(4, 5, 8) > 90:
                gesture = "Eight"
            elif fingers[0] == fingers[1] == 1 and self.get_dist(self.lmList[4]
[1:], self.lmList[8][1:]) < 50:
                gesture = "Heart_single"
            else:
                gesture = "Two"
        elif fingers.count(1) == 5:
            gesture = "Five"
        if self.get_dist(self.lmList[4][1:], self.lmList[8][1:]) < 60 and \
                self.get_dist(self.lmList[4][1:], self.lmList[12][1:]) < 60 and
\
                self.get_dist(self.lmList[4][1:], self.lmList[16][1:]) < 60 and
\
                self.get_dist(self.lmList[4][1:], self.lmList[20][1:]) < 60:
            gesture = "Seven"
        if self.lmList[self.tipIds[0]][2] < self.lmList[self.tipIds[1]][2] and \
                self.lmList[self.tipIds[0]][2] < self.lmList[self.tipIds[2]][2]
and \
                self.lmList[self.tipIds[0]][2] < self.lmList[self.tipIds[3]][2]
and \
```

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                self.lmList[self.tipIds[0]][2] < self.lmList[self.tipIds[4]][2]
and \
                self.calc_angle(self.tipIds[1] - 1, self.tipIds[1] - 2,
self.tipIds[1] - 3) > 150.0:
            gesture = "Eight"
        return gesture

class HandGestureNode(Node):
    def __init__(self):
        super().__init__('hand_gesture_node')
        self.publisher_ = self.create_publisher(Image, 'hand_gesture_image', 10)
        self.timer = self.create_timer(0.1, self.timer_callback)
        self.bridge = CvBridge()
        self.hand_detector = HandDetector(detectorCon=0.75)
        self.capture = cv.VideoCapture(0, cv.CAP_V4L2)
        self.capture.set(cv.CAP_PROP_FRAME_WIDTH, 640)
        self.capture.set(cv.CAP_PROP_FRAME_HEIGHT, 480)
        self.pTime = 0

    def timer_callback(self):
        ret, frame = self.capture.read()
        if not ret:
            self.get_logger().error('Failed to capture frame')
            return

        frame, img = self.hand_detector.findHands(frame, draw=False)
        if len(self.hand_detector.lmList) != 0:
            totalFingers = self.hand_detector.get_gesture()
            cv.rectangle(frame, (0, 430), (230, 480), (0, 255, 0), cv.FILLED)
            cv.putText(frame, str(totalFingers), (10, 470),
cv.FONT_HERSHEY_PLAIN, 2, (255, 0, 0), 2)

        if cv.waitKey(1) & 0xFF == ord('q'):
            self.capture.release()
            cv.destroyAllWindows()
            rclpy.shutdown()
            return

        cTime = time.time()
        fps = 1 / (cTime - self.pTime)
        self.pTime = cTime
        text = "FPS : " + str(int(fps))
        cv.putText(frame, text, (10, 30), cv.FONT_HERSHEY_SIMPLEX, 0.9, (0, 0,
255), 1)
        dist = self.hand_detector.frame_combine(frame, img)
        cv.imshow('dist', dist)

        msg = self.bridge.cv2_to_imgmsg(frame, "bgr8")
        self.publisher_.publish(msg)

def main(args=None):
    rclpy.init(args=args)
    hand_gesture_node = HandGestureNode()
    rclpy.spin(hand_gesture_node)
    hand_gesture_node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
```

```
    main()
```